

Advanced Dynare

Chapter IV

Stochastic simulations at first order

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Outline

1. A Basic RBC Model

1.1 Household Problem

1.2 The Firm Problem

1.3 Equilibrium

2. Solving the Model

2.1 Local Solutions by Linear Approximation

2.2 Generic Representation of the Linearized RE-Model

2.3 Solving Difference Equations

2.4 The Blanchard-Kahn Conditions

2.5 Solution via Eigenvalue Decomposition

3. Dynare implementation

Dynamic Stochastic General Equilibrium models

- Most macro models belong to larger class of **DSGE** models: intertemporal and thus **dynamic**
- Economy is subject to shocks: **stochastic**
- Goods, capital and labor market interact and only few variables are left exogenous: **general equilibrium** models
- We consider a simple RBC model

Problem setup: consumer problem

- The HH maximizes the utility function

$$\max_{\{c_t, k_{t+1}, l_t\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{c_t^{1-\sigma} - 1}{1-\sigma} + \psi \log(1 - l_t) \right) \quad (1)$$

- subject to the budget constraint

$$c_t + i_t = w_t l_t + R_t k_{t-1} \quad \forall t, \quad (2)$$

where c_t is consumption, i_t investment, l_t labor supply, k_t capital stock, w_t real wage, and R_t real rental rate of capital

- Setup assumes decentralized factor markets with competitive firms
- Households own the capital stock and rent it out to firms

Problem setup: laws of motion for capital and TFP

- The second constraint is the law of motion for capital

$$k_t = (1 - \delta)k_{t-1} + i_t \quad \forall t, \quad (3)$$

- Also given: an initial value for capital: $k_{-1} > 0$ and the transversality constraint
- Capital stock timing follows Dynare convention
- Additionally, the HH takes into account the stochastic LOM for z_t

$$z_t = \rho_z z_{t-1} + \varepsilon_t, \varepsilon_t \stackrel{iid}{\sim} (0, \sigma_\varepsilon^2) \quad (4)$$

- Due to our assumptions, we will not have corner solutions and can ignore the non-negativity constraints

The first order conditions

- The first order conditions can be written to yield

$$c_t^{-\sigma} = \beta \mathbb{E}_t c_{t+1}^{-\sigma} ((1 - \delta) + R_{t+1}) \equiv \beta \mathbb{E}_t c_{t+1}^{-\sigma} (1 + r_{t+1}) \quad (5)$$

$$\psi \frac{1}{1 - l_t} = c_t^{-\sigma} w_t \quad (6)$$

- (5) is the familiar Euler equation
- (6) is labor supply equation, indicating amount of labor supplied for given real wage

Firm problem

- We assume a competitive representative firm
- The firm problem is static:

$$\max_{\{l_t, k_{t-1}\}} e^{z_t} k_{t-1}^\alpha l_t^{1-\alpha} - w_t l_t - R_t k_{t-1} \quad (7)$$

- The resulting first order conditions again state that factors are paid their marginal products

$$w_t = (1 - \alpha) e^{z_t} \left(\frac{k_{t-1}}{l_t} \right)^\alpha \quad (8)$$

$$R_t = \alpha e^{z_t} \left(\frac{k_{t-1}}{l_t} \right)^{\alpha-1} \quad (9)$$

Characterizing the solution

- Imposing market clearing, equilibrium can be characterized by two equations:

- The static labor market clearing condition

$$(1 - \alpha)e^{z_t} \left(\frac{k_{t-1}}{l_t} \right)^\alpha = \psi \frac{1}{1 - l_t} \left[e^{z_t} k_{t-1}^\alpha l_t^{1-\alpha} - k_t + (1 - \delta)k_{t-1} \right]^\sigma \quad (10)$$

- The dynamic equation resulting from goods and capital market clearing:

$$\begin{aligned} & \left[e^{z_t} k_{t-1}^\alpha l_t^{1-\alpha} - k_t + (1 - \delta)k_{t-1} \right]^{-\sigma} = \\ & \beta \mathbb{E}_t \left\{ \begin{aligned} & \left[e^{z_{t+1}} k_t^\alpha l_{t+1}^{1-\alpha} - k_{t+1} + (1 - \delta)k_t \right]^{-\sigma} \\ & \left(\alpha e^{z_{t+1}} \left(\frac{k_t}{l_{t+1}} \right)^{\alpha-1} + (1 - \delta) \right) \end{aligned} \right\} \end{aligned} \quad (11)$$

- Solving (10) for l_t and plugging into (11) results in a **second-order difference equation** that, together with the exogenous LOM and the TV/No-Ponzi-condition, characterizes the solution

Equilibrium

- We are looking for a solution to a generic problem of the form

$$0 = \mathbb{E}_t[f(y_{t+1}, y_t, y_{t-1}, \varepsilon_t)] \quad (12)$$

where $y = c_t, l_t, k_{t+1}$

- An equilibrium is an allocation $\{c_t, l_t, k_{t+1}\}_{t=0}^{\infty}$ where
 1. given the exogenous state variable $\{z_t\}_{t=0}^{\infty}$ (4) and k_0
 2. HH maximize their utility ((5) and (6) plus the TVC)
 3. Firms maximize their profits ((9) and (8))
 4. Markets clear
- Problem: These above equations only describe solution implicitly

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Solving the model: The problem

- We have a system with the same number of equations as unknowns
- There are two problems:
 1. The equations are nonlinear
 2. The system is dynamic; the Euler equation is forward-looking, the LOM of capital backward looking
- The second problem can be overcome by mathematical methods (e.g. analytically in the Brock-Mirman model), the first problem is more difficult
- If the system is not convertible to a linear one, we do not know much about its solution
- There are two ways out
 1. Leave the system non-linear, and let the computer solve it (**numerical solutions**)
 2. **Linearize** the system, i.e. take linear approximations that are 'good enough' when the variables are close to the steady-state.

Solution concepts

- Solution to RBC model is characterized by stochastic higher order non-linear difference equation
- Would like explicit solution
→ given state of nature, we would like to directly compute the control variables
- Solution is in principle “**global**”, i.e. policy function $h(x_t)$ specified over the entire state space
- However, obtaining a global solution is often difficult
- Idea: instead approximate complicated problem with an easier one that is “**locally**” accurate
→ rely on local approximation of $h(x_t)$ around particular point of state space

Locally approximate solution

- We will use a first-order Taylor approximation:
 - Robust against curse of dimensionality
 - Often a good compromise between speed and accuracy
 - Locally very accurate, provided
 - **well-defined approximation point**
 - policy function is **sufficiently smooth** in the approximation point
- Point of approximation: **Deterministic Steady State**, i.e. resting point of the system without shocks for $t \rightarrow \infty$ (and where agents take the absence of uncertainty into account)
- Denote steady state values without subscripts: c, k , etc. and percentage deviations from steady state

$$\hat{x}_t = \frac{x_t - x}{x} \quad (13)$$

Linearized equations

- The resource constraint in linearized form is:

$$\hat{k}_t = \left[\alpha \frac{y}{k} + (1 - \delta) \right] \hat{k}_{t-1} + \frac{y}{k} \left(z_t + (1 - \alpha) \hat{l}_t \right) - \frac{c}{k} \hat{c}_t \quad (14)$$

- Linearization of the Euler equation yields:

$$\hat{c}_t = \mathbb{E}_t \left[\hat{c}_{t+1} - \frac{\beta}{\sigma} \alpha A^* \left(\frac{k}{l} \right)^{\alpha-1} \left(z_{t+1} + (\alpha - 1) \hat{k}_t - (\alpha - 1) \hat{l}_{t+1} \right) \right] \quad (15)$$

- The labor market condition can be approximated as

$$\hat{l}_t = \left(\frac{l}{1-l} + \alpha \right)^{-1} \left(z_t + \alpha \hat{k}_{t-1} - \sigma \hat{c}_t \right) \equiv \gamma_l \left(z_t + \alpha \hat{k}_{t-1} - \sigma \hat{c}_t \right) \quad (16)$$

Linearized equations: Substituting out l_t

- Plugging into the Euler equation (15) yields:

$$\hat{c}_t = \mathbb{E}_t \left[\underbrace{-\frac{\beta}{\sigma} \alpha A^* \left(\frac{k}{l}\right)^{\alpha-1} (\alpha-1) (1-\gamma_l \alpha) \hat{k}_t}_{\alpha_4} + \underbrace{\left(1 - \frac{\beta}{\sigma} \alpha A^* \left(\frac{k}{l}\right)^{\alpha-1} (\alpha-1) \gamma_l \sigma\right)}_{\alpha_5} \hat{c}_{t+1} + \underbrace{-\frac{\beta}{\sigma} \alpha A^* \left(\frac{k}{l}\right)^{\alpha-1} (1 - (\alpha-1) \gamma_l) z_{t+1}}_{\alpha_6} \right] \quad (17)$$

Linearized equations: Substituting out l_t

- Plugging into the resource constraint (14) results in

$$\begin{aligned}\hat{k}_t = & \underbrace{\left(\alpha \frac{y}{k} (1 + (1 - \alpha) \gamma_l) + (1 - \delta) \right)}_{\alpha_1} \hat{k}_{t-1} \\ & + \underbrace{- \left(\frac{c}{k} + \frac{y}{k} (1 - \alpha) \gamma_l \sigma \right)}_{\alpha_2} \hat{c}_t \\ & + \underbrace{\left(\frac{y}{k} (1 + (1 - \alpha) \gamma_l) \right)}_{\alpha_3} z_t\end{aligned}\tag{18}$$

Finding the rational-expectations equilibrium

- Having linearized the equations (17) and (18), we have solved number one of the two problems:
 1. The equations are nonlinear
 2. The system is dynamic; the Euler equation is forward-looking, the LOM of capital backward looking
- We are looking for decisions rules of the form:

$$\hat{k}_t = \phi_{kk}\hat{k}_{t-1} + \phi_{kz}z_t \quad (19)$$

$$\hat{c}_t = \phi_{ck}\hat{k}_{t-1} + \phi_{cz}z_t \quad (20)$$

Numerical structure of the problem

- Consider generic representation of a linearized DSGE model like (17) and (18):

$$0 = A\mathbb{E}_t(y_{t+1}) + By_t + Cy_{t-1} + D\varepsilon_t \quad (21)$$

with generic linear solution

$$y_t = Fy_{t-1} + G\varepsilon_t \quad (22)$$

- Plug in solution repeatedly to get

$$0 = A\mathbb{E}_t(y_{t+1}) + By_t + Cy_{t-1} + D\varepsilon_t \quad (23)$$

$$= A\mathbb{E}_t(F(Fy_{t-1} + G\varepsilon_t) + G\varepsilon_{t+1}) + B(Fy_{t-1} + G\varepsilon_t) + Cy_{t-1} + D\varepsilon_t \quad (24)$$

$$= \underbrace{\left[AF^2 + BF + C \right]}_{\stackrel{!}{=}0} y_{t-1} + \underbrace{\left[(AF + G)G + D \right]}_{\stackrel{!}{=}0} \varepsilon_t \quad (25)$$

- We need to solve a quadratic matrix equation (and select the correct root)!

Finding the rational-expectations equilibrium

- There exist various solution approaches to solve this **Ricatti equation**
 1. Use diagonalization of the transition matrix (Blanchard and Kahn 1980)
→ done in the following for didactical reasons, but requires substituting out static variables like labor
 2. More sophisticated and numerically stable version based on generalized Schur decomposition (see Klein (2000))
→ default in Dynare (see Villemot 2011)
 3. Iterative procedure called “Cyclic reduction” (Bini et al. 2002) that is often faster on big models
→ available in Dynare `dr=cycle_reduction` option
 4. Using Newton solver (e.g. Meyer-Gohde and Saecker 2024)
→ abandoned a long time ago in Dynare but maybe worth revisiting

Generic representation of linear equilibrium conditions

- Denote the vector of control variables with u_t and the state vector with $x_t = [x_{t-1}^1; x_t^2]'$, with x_t^2 being a vector of exogenous states:

$$x_t^2 = \Gamma x_{t-1}^2 + \epsilon_t, \quad (26)$$

where ϵ_t is a vector of i.i.d. innovations with bounded support, mean zero, and variance/covariance matrix Σ

- After linearizing all equilibrium conditions and applying the expectation operator, we obtained a linear system of difference equations of the type

$$A\mathbb{E}_t a_{t+1} = B a_t, \quad (27)$$

where $a_t = \begin{bmatrix} \hat{x}_t &; \hat{u}_t \\ (1 \times n_x) & (1 \times n_u) \end{bmatrix}'$

- Note: $\mathbb{E}_t \hat{x}_t^1 = \hat{x}_t^1$ (endogenous state variables are in the information set)
- Solve system (27), while ruling out explosive paths

$$\lim_{t \rightarrow \infty} |a_t| < \infty \quad (28)$$

Generic representation of linear equilibrium conditions: RBC

- Our RBC example can be written as:

$$\begin{bmatrix} 1 & 0 & 0 \\ \alpha_4 & \alpha_6 & \alpha_5 \\ 0 & 1 & 0 \end{bmatrix} \mathbb{E}_t \begin{pmatrix} \hat{k}_t \\ z_{t+1} \\ \hat{c}_{t+1} \end{pmatrix} = \begin{bmatrix} \alpha_1 & \alpha_3 & \alpha_2 \\ 0 & 0 & 1 \\ 0 & \rho_z & 0 \end{bmatrix} \begin{pmatrix} \hat{k}_{t-1} \\ z_t \\ \hat{c}_t \end{pmatrix}$$

Digression: Difference equations

- Before solving the whole matrix equation (27), we first explore **univariate difference equations**
- Key distinction
 - Pre-determined variables (such as state variables)
 - Control variables (or forward-looking variables)
- Introduce **generic notation** for the univariate case:
 - y_t now denotes the endogenous variable
 - ϵ_t denotes an exogenous variable
- Both are scalars
- A **solution** relates y_t to past, present, and future values of ϵ_t (and a boundary condition)

Pre-determined variables

- Assume y_t is a state variable determined by

$$y_t = ay_{t-1} + \epsilon_t \quad (29)$$

- Solve this equation by backward iteration

$$y_t = a(ay_{t-2} + \epsilon_{t-1}) + \epsilon_t \quad (30)$$

$$= a^t y_0 + a^{t-1} \epsilon_1 + a^{t-2} \epsilon_2 + \dots + a \epsilon_{t-1} + \epsilon_t. \quad (31)$$

- y_t is a function of the realizations of ϵ_s for $s = 1, \dots, t$ and initial condition y_0
- With predetermined variables, we have a stable solution only if (we have an initial condition and) $|a| < 1$. If $|a| > 1$ there is no stable solution

Control/Forward-looking variables I

- Assume y_t is a control variable determined by (e.g. an Euler equation)

$$ay_t = \mathbb{E}_t y_{t+1} + \epsilon_t \quad (32)$$

- Again solution depends on a
- Case $|a| > 1$: Rewrite and iterate forward

$$\begin{aligned} y_t &= \frac{1}{a} \mathbb{E}_t y_{t+1} + \frac{1}{a} \epsilon_t \\ y_t &= \underbrace{\lim_{k \rightarrow \infty} \left(\frac{1}{a} \right)^{k+1} \mathbb{E}_t (y_{t+k+1})}_{=0 \text{ by (28) and } a > 1} + \sum_{i=0}^{\infty} \left(\frac{1}{a} \right)^{i+1} \mathbb{E}_t \epsilon_{t+i} \end{aligned} \quad (33)$$

Control/Forward-looking variables II

- Case $|a| < 1$: forward-iteration of (32) does not yield stable solution
- Compute backward-solution; define expectational error: $\xi_{t+1} = y_{t+1} - \mathbb{E}_t y_{t+1}$, and rewrite (32):

$$ay_t = y_{t+1} - \xi_{t+1} + \epsilon_t \quad (34)$$

$$\text{or: } y_t = ay_{t-1} + \xi_t - \epsilon_{t-1} \quad (35)$$

- Solving backward until

$$y_t = a^t y_0 + \sum_{s=1}^t a^{t-s} \xi_s - \sum_{s=1}^t a^{t-s} \epsilon_{s-1}, \quad (36)$$

which is a **stable solution** for $|a| < 1$.

- Note that it **holds for any realization of the sequence of expectational errors** $\{\xi_s\}_{s=1\dots t}$, i.e. there is no unique solution (indeterminacy)

Univariate case: summary

- We need an explosive autoregressive coefficient ($|a| > 1$) when dealing with a forward-looking variable to obtain a unique and stable solution
- We need a stable autoregressive coefficient ($|a| < 1$) when dealing with a backward looking variable to obtain a stable solution (initial condition makes it unique)

Blanchard-Kahn conditions

- Back to the matrix equation case
- Write linearized generic DSGE model (27) as

$$\mathbb{E}_t a_{t+1} = \underbrace{A^{-1}B}_{\equiv W} a_t \quad (37)$$

- Inverting A requires the **rank condition** to be satisfied
- Solution to (37) depends on the eigenvalues of W , as established by Blanchard and Kahn (1980)
 - If the number of eigenvalues of W outside the unit circle (explosive) is equal to the number of non-predetermined (i.e. forward-looking) variables, then there exists a **unique bounded solution**
 - If number of eigenvalues of W outside the unit circle (explosive) exceeds number of forward-looking variables: **no stable solution**
 - If number of eigenvalues of W outside the unit circle (explosive) less than number of forward-looking variables: **infinity of solutions**

Classifying variables: `model_info`

`RBC_IRF.mod`

- To display the variable classification in Dynare, use

```
model_info;
```

```
Information about RBC_IRF (dynamic model)
```

```
The following variables appear with a lag and are therefore states:
```

```
z(-1)
```

```
log(k(-1)) (original expression log(k))
```

```
The following variables appear with a lead and are therefore forward-looking variables:
```

```
z(1)
```

```
log(c(1)) (original expression log(c))
```

```
log(l(1)) (original expression log(l))
```

```
The following variables do not appear with a lead or lag and are therefore purely static variables:
```

```
y
```

```
c
```

```
k
```

```
l
```

```
w
```

```
invest
```

```
r
```

```
log(y()) (original expression log(y))
```

```
log(w()) (original expression log(w))
```

```
log(invest()) (original expression log(invest))
```

Checking Blanchard/Kahn conditions

RBC_IRF.mod

- To check the Blanchard/Kahn conditions, use

```
check;
```

EIGENVALUES:

Modulus	Real	Imaginary
0.9531	0.9531	0
0.97	0.97	0
1.057	1.057	0
3.108e+16	-3.108e+16	0
4.196e+32	4.196e+32	0

There are 3 eigenvalue(s) larger than 1 in modulus for 3 forward-looking variable(s)
The rank condition is verified.

Solving the system: Eigenvalue decomposition

- Recall that a_t in (37) contains two state (backward-looking) variables: $x_t \equiv [\hat{k}_{t-1} \ z_t]'$ and one control (forward-looking) variable $u_t = \hat{c}_t$
- Decouple the system using an **Eigenvalue (Jordan) decomposition**:

$$W = D\Lambda D^{-1}, \quad (38)$$

where matrix Λ has eigenvalues on its main diagonal and D is a matrix of eigenvectors

- Then system (37), $\mathbb{E}_t a_{t+1} = W a_t$, can be rewritten as

$$\begin{aligned} \mathbb{E}_t \underbrace{D^{-1} a_{t+1}}_{\equiv \zeta_{t+1}} &= \Lambda \underbrace{D^{-1} a_t}_{\equiv \zeta_t} \\ \mathbb{E}_t \zeta_{t+1} &= \Lambda \zeta_t \end{aligned} \quad (39)$$

- This yields a set of independent univariate difference equations in $\zeta_{i,t}$

$$\begin{aligned} \mathbb{E}_t \zeta_{1,t+1} &= \lambda_1 \zeta_{1,t} \\ &\vdots \\ \mathbb{E}_t \zeta_{i,t+1} &= \lambda_i \zeta_{i,t} \end{aligned}$$

Solving the system: Reordering

- Elements in Λ can be ordered in ascending order without loss of generality (note: we need to reshuffle the eigenvectors accordingly)
- Assume that number of explosive eigenvalues equal to n_u and partition reordered D and Λ and matrix W according to stable and explosive eigenvalues

$$\Lambda = \begin{bmatrix} \Lambda_1 & 0 \\ n_x \times n_x & n_x \times n_u \\ 0 & \Lambda_2 \\ n_u \times n_x & n_u \times n_u \end{bmatrix}; W = \begin{bmatrix} w_{11} & w_{12} \\ n_x \times n_x & n_x \times n_u \\ w_{21} & w_{22} \\ n_u \times n_x & n_u \times n_u \end{bmatrix}; D^{-1} = \begin{bmatrix} q_{11} & q_{12} \\ n_x \times n_x & n_x \times n_u \\ q_{21} & q_{22} \\ n_u \times n_x & n_u \times n_u \end{bmatrix}$$

- Also partition $\zeta_t = \begin{bmatrix} \zeta_{1,t} & \zeta_{2,t} \\ 1 \times n_x & 1 \times n_u \end{bmatrix}'$

Solving the system: Solution for controls

- First, consider the lower part of (39) corresponding to controls: $\mathbb{E}_t \zeta_{2,t+1} = \Lambda_2 \zeta_{2,t}$
- As in the univariate case, this part is iterated forward

$$\zeta_{2,t} = \lim_{k \rightarrow \infty} \Lambda_2^{-k} \mathbb{E}_t \zeta_{2,t+k} = 0, \quad (40)$$

as the diagonal elements of Λ_2 are explosive and $\lim_{t \rightarrow \infty} |\zeta_{2,t}| < \infty$ (linear combination of bounded variables)

- From the definition of ζ_t follows

$$\begin{bmatrix} \zeta_{1,t} \\ n_x \times 1 \\ \zeta_{2,t} \\ n_u \times 1 \end{bmatrix} = \underbrace{\begin{bmatrix} q_{11} & q_{12} \\ n_x \times n_x & n_x \times n_u \\ q_{21} & q_{22} \\ n_u \times n_x & n_u \times n_u \end{bmatrix}}_{D^{-1}} \underbrace{\begin{bmatrix} \hat{x}_t \\ \hat{u}_t \end{bmatrix}}_{a_t} \quad (41)$$

- Using $\zeta_{2,t} = 0$ implies we have a linear policy function for the controls:

$$0 = q_{21} \hat{x}_t + q_{22} \hat{u}_t \Rightarrow \hat{u}_t = \underbrace{-q_{22}^{-1} q_{21}}_{\equiv M} \hat{x}_t \quad (42)$$

Solving the system: Solution for states I

- Given the partitioning of W , the upper part of

$$\mathbb{E}_t a_{t+1} = \underbrace{A^{-1}B}_{\equiv W} a_t \quad (37)$$

is given

$$\mathbb{E}_t \hat{x}_{t+1} = w_{11} \hat{x}_t + w_{12} \hat{u}_t, \quad (43)$$

- Substituting for $\hat{u}_t$ using

$$\hat{u}_t = \underbrace{-q_{22}^{-1} q_{21}}_{\equiv H} \hat{x}_t = H \hat{x}_t \quad (42')$$

gives

$$\mathbb{E}_t \hat{x}_{t+1} = \underbrace{(w_{11} - w_{12} q_{22}^{-1} q_{21})}_{\equiv F} \hat{x}_t \quad (44)$$

Solving the system: Solution for states II

- Denote the number of endogenous states with n_{x1} and of the exogenous states with n_{x2}
- With endogenous state variables being predetermined

$$\hat{x}_t^1 = \mathbb{E}_t \hat{x}_t^1 \quad (45)$$

and given the exogenous LOM

$$x_t^2 = \Gamma x_{t-1}^2 + \epsilon_t, \quad (26)$$

we have

$$\hat{x}_{t+1} = \mathbb{E}_t \hat{x}_{t+1} + \underbrace{\begin{bmatrix} 0_{n_{x1} \times n_{x2}} \\ I_{n_{x2} \times n_{x2}} \end{bmatrix}}_{\equiv \eta} \epsilon_{t+1} = F \hat{x}_t + \eta \epsilon_{t+1} \quad (46)$$

The solution

- Taken together, the solution equations (42) and (46) provide a recursive representation of the solution to (27) in **state space form**:

$$\hat{u}_t = H\hat{x}_t \quad (47)$$

$$\hat{x}_{t+1} = F\hat{x}_t + \eta\epsilon_{t+1} \quad (48)$$

- The first equation is called **observation equation** or **measurement equation**, while the second one is called **state transition equation**

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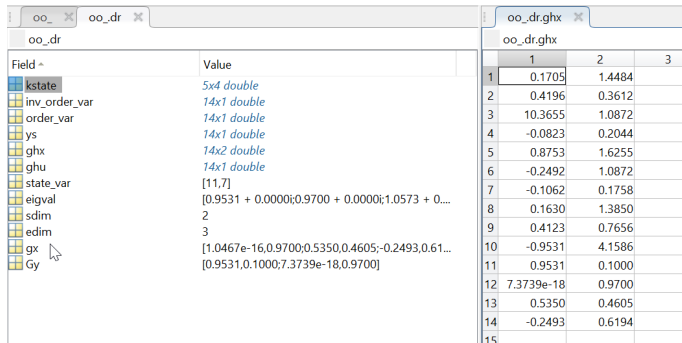
3. Dynare implementation

Solution storage in Dynare

- Note that Dynare internally stacks controls and states into one equation:

$$y_t = y + gh_x(x_{t-1} - x) + gh_u\epsilon_t \quad (49)$$

where gh_x and gh_u are subfields of oo_dr



The screenshot displays the Dynare command window with two panes. The left pane shows the structure of the `oo_dr` variable, listing fields and their dimensions. The right pane shows the numerical values for the `oo_dr.ghx` subfield, organized in a table with 15 rows and 3 columns.

Field	Value
<code>kstate</code>	<code>5x4 double</code>
<code>inv_order_var</code>	<code>14x1 double</code>
<code>order_var</code>	<code>14x1 double</code>
<code>ys</code>	<code>14x1 double</code>
<code>ghx</code>	<code>14x2 double</code>
<code>ghu</code>	<code>14x1 double</code>
<code>state_var</code>	<code>[11,7]</code>
<code>eigval</code>	<code>[0.9531 + 0.0000i;0.9700 + 0.0000i;1.0573 + 0....</code>
<code>sdim</code>	<code>2</code>
<code>edim</code>	<code>3</code>
<code>gx</code>	<code>[1.0467e-16,0.9700;0.5350,0.4605;-0.2493,0.61...</code>
<code>Gy</code>	<code>[0.9531,0.1000;7.3739e-18,0.9700]</code>

	1	2	3
1	0.1705	1.4484	
2	0.4196	0.3612	
3	10.3655	1.0872	
4	-0.0823	0.2044	
5	0.8753	1.6255	
6	-0.2492	1.0872	
7	-0.1062	0.1758	
8	0.1630	1.3850	
9	0.4123	0.7656	
10	-0.9531	4.1586	
11	0.9531	0.1000	
12	7.3739e-18	0.9700	
13	0.5350	0.4605	
14	-0.2493	0.6194	
15			

Decision rules in Dynare

RBC_IRF.mod

- Dynare will display the decision rules in digestible form at order<3

```
stoch_simul(order=1);
```

POLICY AND TRANSITION FUNCTIONS

	LOG_y	LOG_invest	LOG_c	LOG_l	LOG_w
Constant	0.044764	-1.341530	-0.242918	-1.108663	0.752949
log(k(-1))	0.162999	-0.953066	0.535021	-0.249255	0.412254
z(-1)	1.385019	4.158591	0.460495	0.619431	0.765588
eps_z	1.427855	4.287208	0.474737	0.638589	0.789266

- The results are stored in oo_.dr

Making use of the solution

RBC_baseline.mod

- Well-established techniques for linear state space solutions allow computation of theoretical moments, even with linear filters, and of (conditional) variance decompositions

```
stoch_simul(order=1,irf=20,conditional_variance_decomposition=[1,4]);  
stoch_simul(order=1,hp_filter=1600);
```

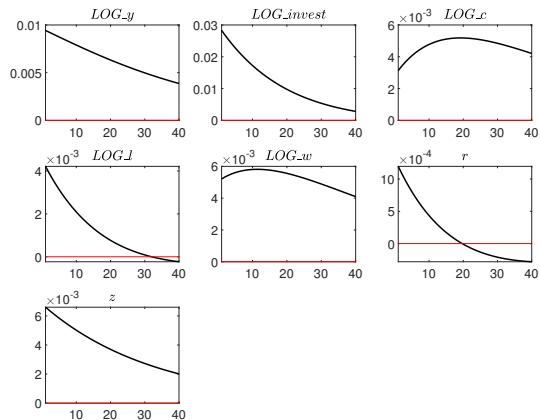
- Given x_0 , state-space solution can be used to simulate equilibrium time series for given sequence $\{\epsilon_t\}_{t=0}^{\infty}$
→ allows computation of simulated moments, impulse responses, conditional and unconditional forecasts

```
stoch_simul(order=1,irf=20,periods=1000,drop=200);
```

- At order=1, IRFs are invariant to the point in the state space
→ only one simulation necessary

RBC Output

```
THEORETICAL MOMENTS (HP filter, lambda = 1600)
VARIABLE      MEAN    STD. DEV.    VARIANCE
LOG_y          0.0448    0.0123       0.0002
LOG_invest     -1.3415    0.0369       0.0014
LOG_c          -0.2429    0.0045       0.0000
LOG_l          -1.1087    0.0055       0.0000
LOG_w          0.7529    0.0069       0.0000
r              0.1269    0.0016       0.0000
z              0.0000    0.0086       0.0001
```



RBC Output with government spending shocks

VARIANCE DECOMPOSITION (in percent) (HP filter, lambda = 1600)

	eps_z	eps_g
LOG_y	96.98	3.02
LOG_k	99.52	0.48
LOG_c	83.95	16.05
LOG_l	65.57	34.43
LOG_w	98.26	1.74
r	97.09	2.91
z	100.00	0.00
ghat	0.00	100.00

CONDITIONAL VARIANCE DECOMPOSITION (in percent)

Period 4:

	eps_z	eps_g
LOG_y	96.82	3.18
LOG_k	99.50	0.50
LOG_c	85.50	14.50
LOG_l	58.89	41.11
LOG_w	98.45	1.55
r	96.41	3.59
z	100.00	0.00
ghat	0.00	100.00

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